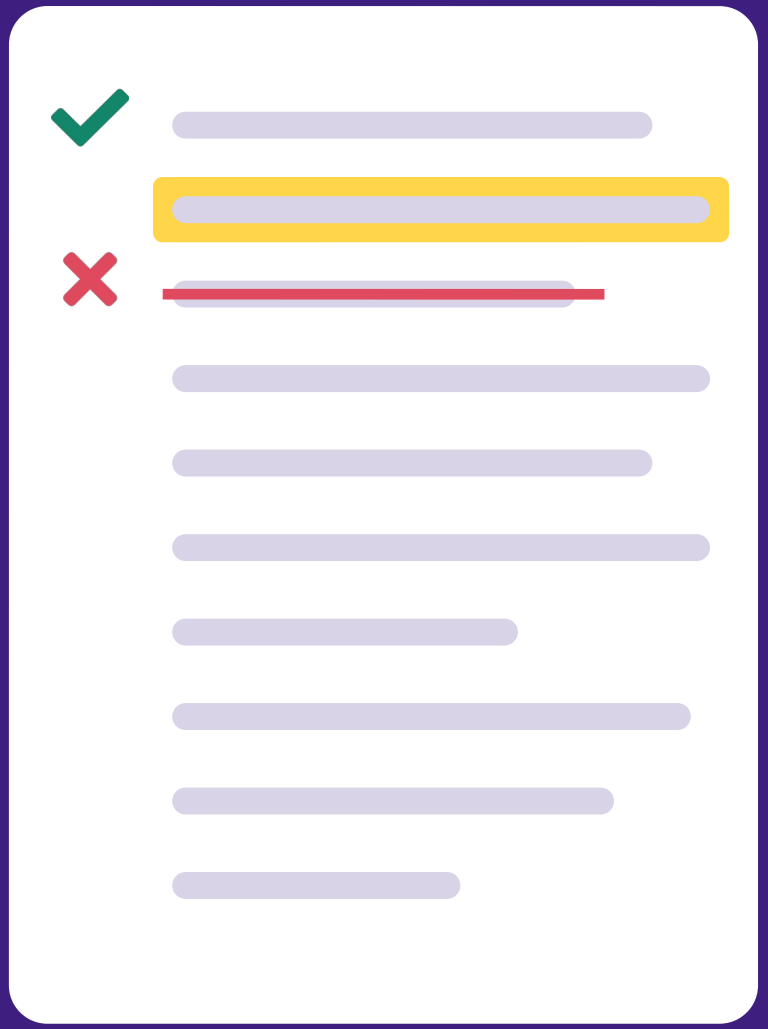
Enhancing Literacy and Numeracy Skills Using AI

AI-Aware Assessment

Designing for integrity, visible thinking and feedback



Ground

What would count as convincing evidence?

Task: Write a short reflection on a practicum lesson, explaining one decision you would change.

Submission A

Reflection: My practicum lesson on persuasive writing went well overall. Learners were engaged and most completed the task. If I taught it again, I would change my questioning so that more learners contribute. Effective questioning supports participation, so I will plan open questions in advance.

Submission B

The same reflection, plus a short record of decisions. Decision: I first wrote that the lesson went well because learners were quiet and finished on time. Revision: after re-reading my observation notes, I changed this, because only four learners spoke and most finished by copying the model. Why this fits my learners: they are confident writers but hesitant speakers, so I will plan paired talk before whole-class questions.

Which submission gives you stronger evidence that the student can actually do what the assessment claims to assess? What evidence made the difference?

Ground

Either, both or neither submission could have involved AI.

“The assessment problem is not simply whether AI was used. The question is whether we can still make a defensible judgement about learning.”

Readiness

Rank the evidence

A

AI is prohibited, but the task is a generic take-home essay submitted only as a polished final product.

B

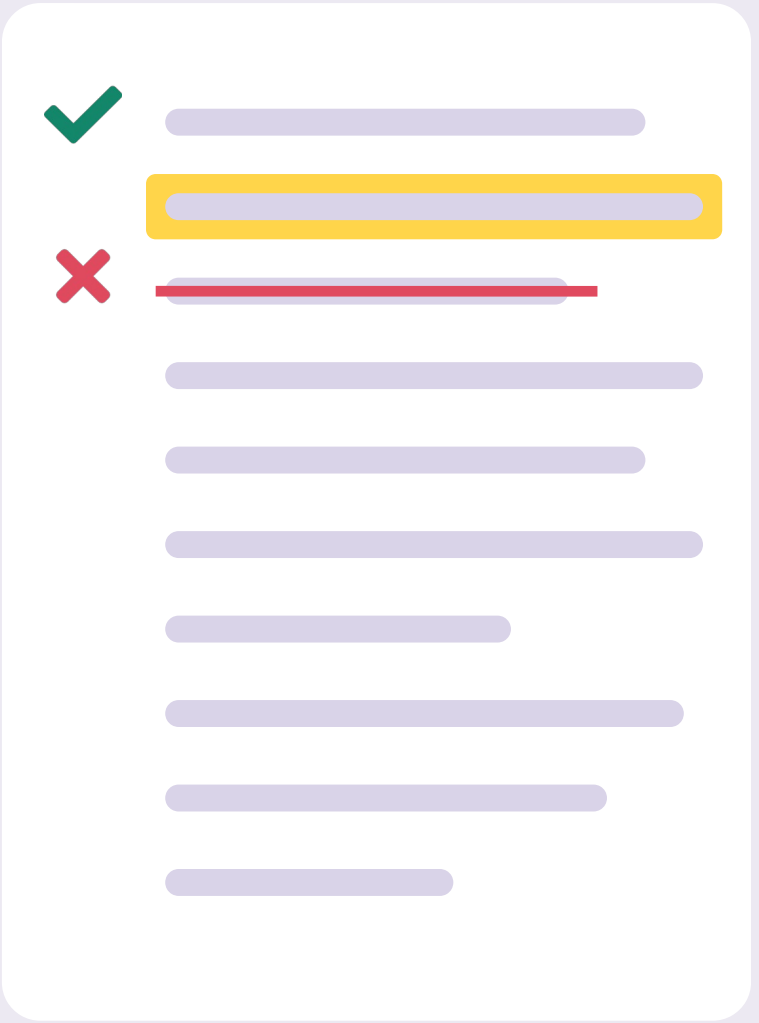
AI is permitted for brainstorming, but students must justify key decisions, show one revision and discuss their reasoning.

C

AI is deliberately integrated into the task, but only the final polished product is assessed.

Rank the three from strongest to weakest evidence of learning.

Assessment integrity is strengthened through design, clarity and visible evidence of learning, not through detection or prohibition alone.



Concept 1

Diagnose an AI-fragile task

PPG assessment: “Write a 1500-word essay explaining three strategies for differentiating instruction. Use at least five references.”

1. What is the intended learning?

2. What could a generative AI tool do easily here?

3. What important student thinking is invisible?

4. What evidence would you trust if you had to judge whether the student could apply this knowledge in practice?

Each group posts one vulnerability to the shared board or chat.

Concept 1

Product plus process

“AI-resistant” does not mean technologically impossible to complete with AI. It means the task generates stronger evidence of the learner’s own reasoning or performance.

The same learning intention could be evidenced through:

- a short case application
- a decision rationale
- staged planning
- an oral explanation
- a revision commentary
- a critique of an AI-generated proposal

Nguyen et al., 2025; Weng et al., 2024

Concept 1

“A polished product is evidence. It is not always enough evidence.”

Concept 2

Redesign for a PPG context

Same underlying assessment, a different context card for each group:

Context A

PPG students preparing for primary practicum.

Context B

PPG students specialising in secondary mathematics.

Context C

Mentor teachers assessing lesson reflection after practicum.

Context D

A mixed-experience PPG cohort completing an online professional task.

Without using AI, redesign the assessment using at least two integrity moves. Preserve the original learning intention and cognitive demand. Then state exactly what new evidence of thinking becomes visible.

Concept 2

Make AI permissions explicit

No AI

The task or stage requires independent demonstration of knowledge or performance.

AI for preparation

AI may support planning, idea generation or resource finding, but not produce assessed work.

AI-supported

AI may support a defined part of the process, such as language refinement, questions or feedback.

AI-integrated

AI use is deliberately built into the assessed task and students must explain or justify how they used it.

AI as object of critique

Students analyse, verify, correct or improve AI output as part of the disciplinary learning.

Decide the permitted role of AI separately for planning, drafting, feedback, revision, submission and evaluation. For evaluation, final judgement always remains human. Name one stage where you would restrict AI and one where you would permit or deliberately integrate it.

Concept 2

Beyond Detection: Designing AI-Aware Assessment & Integrity

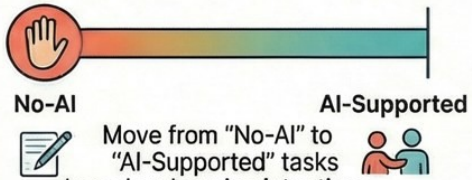
DESIGN-LED INTEGRITY STRATEGIES



Making Thinking Visible

Use staged drafting, process documentation, and oral explanations to ensure authentic student work.

The AI Spectrum Model



Move from "No-AI" to "AI-Supported" tasks based on learning intentions and cognitive demand.



Human-in-the-Loop Authority

AI supports efficiency, but teachers must retain final evaluative authority over student performance.

REDEFINING STUDENT MASTERY



Dual Literacy Requirements

Students require both disciplinary content mastery and the ability to critically evaluate AI outputs.



Critical AI Engagement

Assessment must target the ability to recognize AI bias, hallucinations, and privacy risks.



Formative AI Integration

Use AI to generate feedback prompts and identify misconceptions without replacing professional judgment.

Concept 2

An AI-produced redesign

Original task: "Write a 1500-word essay explaining three strategies for differentiating instruction. Use at least five references."

AI suggestion

1. Submit a 500-word plan explaining your three chosen strategies, with at least three references, in Week 3.
2. Complete a 10-question online quiz on differentiation terminology.
3. Submit your final 1500-word essay with a 200-word reflection on how your ideas changed.
4. Attend a 5-minute oral check-in to discuss your essay.
5. Include your full AI chat history as an appendix.

Concept 2

Critique an AI-produced redesign



What did this improve?



What weakened the assessment or cognitive demand?



What would you reject or revise?

“A clear AI rule is part of assessment design. It should follow the learning intention, not precede it.”

Concept 3

Human feedback and AI feedback

Fictional PPG student response

“I would give the students who are struggling a simpler version of the task so they can finish it successfully. The stronger students can complete the original task and then do an extension activity. This means every student can work at their own level and no one becomes frustrated.”

Learning intention: “Justify a differentiated teaching decision using evidence about learner readiness while preserving the intended learning outcome.”

First, write one formative next-step comment on your own. Then we will use the approved AI tool live, and compare.

Concept 3

Live AI prompt, then compare

Live AI prompt

“Generate three formative questions or next-step prompts that would help this learner improve against the stated learning intention. Do not assign a grade or achievement level.”

What did each notice?

What did either miss?

Which comments would genuinely help the learner progress?

What depends on knowledge of the learner or teaching context?

Concept 3

AI drafts question stems for teacher review.

defensible with oversight

context-dependent

not appropriate for this programme

Concept 3

AI identifies possible misconceptions.

defensible with oversight

context-dependent

not appropriate for this programme

Concept 3

AI drafts formative feedback that the teacher reviews and edits.

defensible with oversight

context-dependent

not appropriate for this programme

Concept 3

AI assigns the final achievement level or grade.

defensible with oversight

context-dependent

not appropriate for this programme

Concept 3

“AI may suggest. The educator evaluates.”

Ilieva et al., 2025

Express

Defend and revise your assessment decision

Return to the assessment you brought from your own practice. Complete six statements:

My assessment is...

The learning I need to see is...

The evidence I will make visible is...

AI may be used for...

AI will not be used for...

Professional judgement remains with...

Express

Pair exchange

One strength

One risk

One question

Each partner responds with one strength, one risk and one question. Then two or three participants share one decision they changed after peer discussion.

“The aim is not to make assessment AI-proof. It is to design assessment that gives us defensible evidence of learning in an AI-aware environment.”

What you carry into your lesson plans

One redesigned PPG assessment or assessment-learning activity containing:

- a clear statement of the intended learning
- a description of the evidence of reasoning, process or performance that will be made visible
- an explicit statement of where AI is prohibited, permitted, supported or integrated
- a clear human evaluative boundary
- one planned formative use of AI that remains under educator review
- a short justification explaining why the redesign strengthens the validity of the assessment

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